1. Problem statement

The subscription business mainly relied on the old customer who renewed. However, customers may differ in their renewal behavior depending on their subscription plans and transaction patterns. Understanding these differences can help businesses identify potential churn risks and develop more effective customer-retention strategies.

This project aims to explore customer subscription renewal behavior and examine whether retention patterns differ between monthly and annual subscribers. We will find out which customer and usage characteristics, such as engagement, plan tier, and subscription duration.Through EDA and visualizations, we hope to identify meaningful patterns associated with customer retention and provide insights that may help subscription-based businesses improve their business strategies.

To keep the project manageable in one semester, we will focus on one single dataset.

2. Data sources

We plan to use the dataset containing customer transaction and subscription records from KKBOX, a music streaming service. The dataset includes variables such as user's id, payment method, subscription plan duration, plan price, actual amount paid, transaction date, membership expiration date, auto renewal status, and cancellation status. These numerical and categorical variables will allow us to explore customer renewal behavior and compare subscription patterns across different plan durations. In particular, we are interested in examining whether subscription duration and payment behavior are associated with customer retention.

We found the dataset on Mendeley Data, a research data repository that provides access to shared datasets. Source: https://data.mendeley.com/datasets/mv3f8bdrvy/1

However, the potential issue is sampling bias and representation. This is a specific dataset, so its users may not represent customers of all subscription-based services. Customer behaviors,

and renewal patterns may differ across platforms, countries, and demographic groups. Therefore, our conclusions should not be generalized to all subscription services.

3. Analysis goals and general methods

The main goal of this project is to understand customer subscription renewal behavior and explore whether renewal patterns differ across subscription plan durations.

- Identify the overall subscription patterns
- Compare subscription duration
- Determine which factors affect subscription duration and explore which factors are most important in explaining it

To achieve the goal, we will use EDA and data visualizations. We will calculate the relative data to compare renewal patterns across subscription plan durations, and use appropriate visualization charts to identify trends and differences. Because renewal status and cancellation status in the dataset are binary variables, we will use logistic regression to predict and examine the relationships between customer transaction characteristics and renewal subscription.

4. **Division of labor**

Lingyi: As a Math and Economics student, I can contribute to developing research questions and hypotheses by explaining why certain variables might be associated with customer renewal and cancellation. Having taken courses in econometrics and machine learning for economics, I can also help analyze regression models and evaluate the relationships between subscription characteristics and cancellation behavior. Additionally, I will help interpret the results, discuss their practical implications, and identify potential limitations from an economics perspective.

Zihan Chen : I can help to search the datasets online. Then clean up and merge different elements in the dataset. Also, I could help with coding and set up a model with my teammates

and handle the statistical tests in order to figure out the relationship between different elements and the churn rate.

Lukas: I can contribute to data preprocessing, cleaning, and transformation to prepare the dataset for analysis. I can also help implement regression models, evaluate their performance, and optimize the analysis workflow. Additionally, I will contribute to identifying suitable technical approaches and ensuring the accuracy and efficiency of our data analysis.